

Daily Geometry Drill #6

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: equal angles standing on the same chord · $PA \cdot PB$ products · Ptolemy $AC \cdot BD = AB \cdot CD + BC \cdot AD$ · Brahmagupta $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ · Pitot for tangential quadrilaterals.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Spot the chord, then either double the angle at the centre or equalise the angle in the same segment.

- In a circle with centre O , the angle at the circumference subtending arc AB is 20° . Write down the angle at the centre.
- Points P, Q, R, S lie on a circle, and $\angle PQR = 35^\circ$. Write down $\angle PSR$.
- $ABCD$ is a cyclic quadrilateral with $\angle A = 70^\circ$ and $\angle B = 85^\circ$. Write down $\angle C$.
- The tangent at T to a circle makes an angle of 40° with the chord TA . By the alternate segment theorem, write down the angle in the alternate segment subtended by TA .
- Chords AB and CD of a circle intersect inside at P , with $PA = 2$, $PB = 6$ and $PC = 3$. Write down PD .
- A tangent from P touches the circle at T , and a secant through P cuts the circle at A then B . Given $PT = 3$ and $PA = 1$, write down PB .
- AB is a diameter of a circle and P is any other point on the circle. Write down $\angle APB$.
- Two secants from the external point P cut the circle as $P-A-B$ and $P-C-D$, with $PA = 4$, $PB = 6$ and $PC = 2$. Write down PD .
- A tangent from P has length 8, and a secant through P meets the circle at A (nearer) and B , with $PA = 4$. Write down PB .
- In the cyclic quadrilateral $ABCD$, $\angle BAC = 30^\circ$. Write down $\angle BDC$.

Section B Manipulation Drills

(≤50 s each)

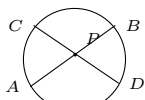
Half the central angle, subtract cyclic opposites from 180° , or multiply chord pieces – then pick from the options.

1. O is the centre of a circle and $\angle AOB = 130^\circ$. The angle at the circumference subtending AB is?
 - A) 60°
 - B) 65°
 - C) 70°
 - D) 130°
2. $ABCD$ is cyclic with $\angle A = 100^\circ$. Then $\angle C$ is?
 - A) 80°
 - B) 100°
 - C) 90°
 - D) 160°
3. Chords AB and CD cross at an interior point P , with $PA = 3$, $PB = 5$ and $PC = 15$. Then PD is?
 - A) 1
 - B) 3
 - C) 5
 - D) 9
4. A tangent from P has length 5, and a secant through P gives outer segment $PA = 2$. Write down PB (the full secant segment).
5. A tangent at T makes an angle of 55° with chord TA . The angle in the alternate segment subtended by TA is?
 - A) 35°
 - B) 55°
 - C) 110°
 - D) 70°
6. The secant PAB from an external point has $PA = 2$, $PB = 18$. The length of the tangent PT drawn from P is?
 - A) 6
 - B) 9
 - C) 12
 - D) 36
7. $ABCD$ is cyclic with $\angle A = 2x$ and $\angle C = 3x$. Find x .
 - A) 24

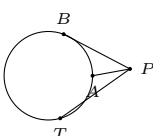
- B) 30
C) 36
D) 72
8. AB is a diameter of a circle and C is a point on the circle with $\angle ABC = 55^\circ$. Write down $\angle BAC$.
9. A *tangential* (circumscribed) quadrilateral $ABCD$ has $AB = 5$, $BC = 7$ and $CD = 6$. By Pitot's theorem ($AB + CD = BC + DA$), find DA .
A) 4
B) 6
C) 8
D) 18
10. Chords AB and CD intersect at P with $PA = 6$, $PB = 4$ and $PC = 3$. Write down PD .

Section C Substitution & Structure(≤ 75 s each)

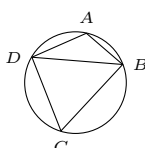
Chain two circle ideas per question: power, Ptolemy and Brahmagupta with a full diagram. All Section C are MCQ.

1.  (after TMUA 2023 Paper 2 Q9) Chords AB and CD meet at P inside a circle with $PA = 6$, $PB = 8$ and $CP : PD = 3 : 4$. Find CP .

- A) 4
B) 6
C) 8
D) 12

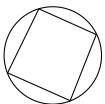
2.  (after TMUA 2017 Paper 1 Q6) From external P , tangent $PT = 12$ touches a circle and secant PAB meets it with $PA = 8$. Find the chord AB .

- A) 6
B) 10
C) 12
D) 18

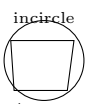
3.  (after TMUA 2017 Paper 2 Q15) Cyclic quadrilateral $ABCD$ has $AB = 6$, $BC = 4$,

$CD = 5$, $DA = 3$ and diagonal $AC = 7$. Find the other diagonal BD (Ptolemy: $AC \cdot BD = AB \cdot CD + BC \cdot AD$).

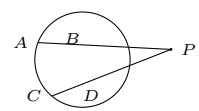
- A) 5
- B) 7
- C) 6
- D) 13

4.  (after BMO1 2010 Q4) A cyclic quadrilateral has sides 4, 5, 5, 6 (in order). By Brahmagupta $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$, its area is?

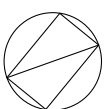
- A) $10\sqrt{6}$
- B) $12\sqrt{5}$
- C) 30
- D) 24

5.  (after BMO1 2015 Q3) A tangential quadrilateral $ABCD$ circumscribes a circle with $AB = 3x$, $BC = 8$, $CD = 5$, $DA = 2x + 3$. Given Pitot $AB + CD = BC + DA$, find x and hence DA .

- A) 12
- B) 15
- C) 18
- D) 9

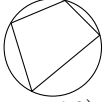
6.  (after TMUA 2019 Paper 1 Q7) From external P , secants PAB ($PA = 4$, $PB = 18$) and PCD ($PC = 6$) cut the same circle. Find CD .

- A) 6
- B) 12
- C) 9
- D) 18

7.  (after TMUA Specimen Paper 2 Q9) Cyclic quadrilateral $ABCD$ has $AB = 8$, $BC = 6$, $CD = 6$, $DA = 6$ and $AC = 12$. Find BD (Ptolemy).

- A) 7
- B) 12
- C) 14

D) 84

8.  (after BMO1 2010 Q4) Cyclic quadrilateral with sides 5, 5, 8, 14 has area (Brahmagupta, $s = 16$) equal to?

A) 44

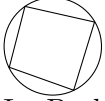
B) 36

C) 48

D) 24

Section D Challenge Ramp(≤ 90 s each)

Hardest 5: stack power, Ptolemy and Brahmagupta – each answer is a single clean integer or surd.

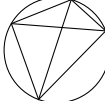
1.  (after BMO1 2010 Q4) Cyclic quadrilateral $ABCD$ has sides 2, 5, 10, 11 (in order, $s = 14$). Its Brahmagupta area is?

A) 36

B) 44

C) 30

D) 48

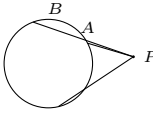
2.  (after BMO1 2011 Q2) Cyclic quadrilateral $ABCD$ has $AB = 7$, $BC = 8$, $CD = 7$, $DA = 9$ with $AC = 11$. Find BD via Ptolemy.

A) 11

B) 7

C) 14

D) 18

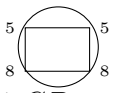
3.  (after TMUA 2020 Paper 1 Q7) From external P , tangent $PT = 12$ and secant PAB satisfies $PB = PA + 10$ with $PT^2 = PA \cdot PB$. Find PA .

A) 6

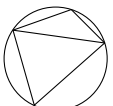
B) 8

C) 12

D) 18

4.  (after *BMO1 2015 Q3*) A quadrilateral with sides 5, 5, 8, 8 (in order $AB = 5, BC = 5, CD = 8, DA = 8$) is both tangential (Pitot $5 + 8 = 5 + 8$) and cyclic. Its Brahmagupta area is?

- A) 40
- B) 30
- C) 36
- D) 44

5.  (after *TMUA 2017 Paper 2 Q15*) Cyclic quadrilateral $ABCD$ has $AB = 6, BC = 8, CD = 12, DA = 10$ and $AC = 8$. Find BD .

- A) 19
- B) 12
- C) 15
- D) 8

“A circle is an equality machine: every theorem just asserts two angles or two products are the same, and your eye should find the chord holding them together.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths